End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech. CSE-AI, Semester:VI
(April -May, 2023)

Subject Code: BAI-304	Subject: Deep Learning-II
Time :3 Hours	Maximum Marks :60

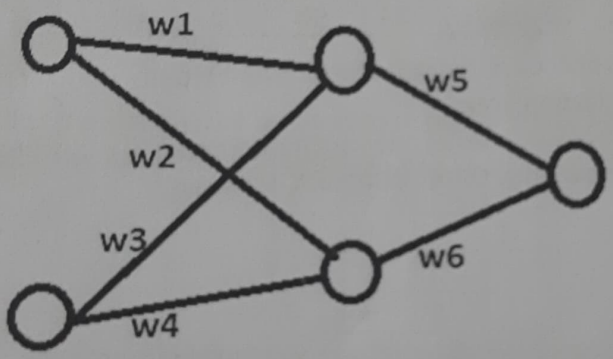
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (2.5*8=20)

a) If CNN filters (no bias) of size 5x5x3 are employed for transforming an image with 3 channels into 23 responses. How many weights need to be learned?
b) How many weights need to be learned on employing a max-pooling kernel of size 4x4 for transforming an image with 3 channels?
c) Identify the problem of using sigmoid function in deep learning models?
d) "After using batch normalization, there is no need to use dropout". Is this statement true? Justify your answer
e) Can classification accuracy be a suitable performance parameter if there are unequal number of data samples in each class? Justify
f) What are the least number of output classes required for drawing confusion matrix?
g) Explain the concept of Transfer learning with an example?
h) Differentiate between Under-complete and Regularized Auto encoders.

UNIT-I

Q2 Consider the given neural network with single input as [4, 5] and output as [7]. The initial weight values (w1, w2, w3, w4, w5, and w6) are [0.2, 0.2, 0.3, 0.1, 0.4, 0.2] and the loss function is mean squared error with learning rate as 0.01. Assuming there are no activation function and biases, answer the following with 2 decimal places: (10)



a) Find the prediction value and error value of the neural network in the first forward pass.
b) Find the updated values of weights (w1 to w6) after back propagation algorithm once.

Q3 Consider a convolution neural network consisting of two convolutional layers and one max-pooling layer. The shape of any matrix is in the form of *channel x height x width.* The size of the input image is 3 x 224 x 224. The first convolutional layer has following configuration: Kernel size: 5x5, No. of kernels: 96, Stride: 1, Padding: 1. The second convolutional layer has following configuration: Kernel size: 6x6, No. of kernels: 128, Stride: 2, Padding: 1. The maxpooling layer has following configuration: Kernel size: 2x2, Stride: 2. Based on the given architecture, answer the following questions: (10)

a) What is the size of the output feature maps after first convolutional layer, second convolutional layer and maxpooling layer?

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b) What is the depth of each convolutional kernel in first and second convolutional layers?

UNIT-II

Q4 a) Which optimization method is used to optimize the learning rate? Explain. **(2.5+ 2.5+5= 10)**

b) What effect does the different values of batch size has on the performance of gradient descent algorithm?

c) Differentiate between rms prop optimizer and Adam optimizer.

Q5 a) Differentiate between L1 and L2 regularization techniques. **(2.5+2.5+5=10)**

b) Is dropout layer active during the testing of a deep neural network? Justify

c) Explain the Batch normalization technique. How is it useful for regularization?

UNIT-III

Q6 **(10)**

	Predicted: No	Predicted: Yes
Actual: No	40	15
Actual: Yes	10	80

Consider the above confusion matrix and calculate the following performance parameters: Accuracy, Error Rate, Precision, Recall, F1-score.

Q7 a) Differentiate between Parameter tuning and Hyper-parameter tuning. **(10)**

b) Explain the following hyper-parameter tuning approaches:

(i) Grid

(ii) Random

UNIT-IV

Q8 Explain the architecture of Auto encoders. Identify the different types of Auto encoders and their application areas. **(10)**

Q9 Discuss the structure of generative adversarial network (GAN) and different applications of GAN? How does GAN distinguish between fake data and real data. **(10)**